BiasGUARRD: Enhancing Fairness and Reliability in LLM Conflict Resolution Through Agentic Debiasing

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Abstract

As foundation models are increasingly deployed in socially sensitive domains, ensuring their reliability in high-stakes, human-centered decision-making is critical. Large language models (LLMs) often mirror human cognitive biases, leading to unfair or inconsistent outcomes. While prior work has identified such biases, we uniquely examine their impact in **interpersonal conflict resolution.** We present:

- A **benchmark of 100 human-annotated neutral conflict scenarios** across family, workplace, community, and friendship contexts.
- A study of LLM susceptibility to **four injected cognitive biases**: affective framing, halo effect, framing effect, and serial order bias; LLMs shift judgment **31-79%** of the time during biased prompts.
- A **novel multi-agent debiasing framework, BiasGUARRD** (**B**ias **G**overnance **U**sing **A**gents for **R**eliable **R**easoning-based **D**ecision-making), that detects and mitigates bias through dynamic reasoning interventions, reducing judgment inconsistency by **up to 63.3%**, outperforming existing bias-reduction strategies.

Introduction

- **LLMs are increasingly used for socially grounded tasks** such as interpersonal conflict resolution, where biased judgments can cause real-world harm.
  - **Conflicts in families, workplaces, and communities** are emotionally complex and impact mental health, relationships, and productivity.
- **Expert-led roleplay** remains the gold standard for conflict training but is time-consuming and costly.
- LLM-based systems offer a scalable alternative grounded in psychological theory but exhibit **behavior analogous to cognitive and social biases**, leading to unfair or sycophantic responses shaped by prompt framing.
  - Prior work studies structured, impersonal tasks, **overlooking emotionally nuanced, ambiguous interpersonal conflicts.**
- We investigate how LLMs handle biased interpersonal conflict scenarios and introduce **BiasGUARRD**, a **multi-agent framework to detect and mitigate cognitive biases in LLM decision-making**.

Methods

Dataset: Interpersonal Conflict Benchmark

- **100 neutral, human-annotated conflict scenarios** between Person A and Person B across diverse (workplace, family, etc.) domains.
  - Generated by **GPT-4o-mini**, designed to avoid bias cues.
- Each prompt ends by requiring the model to **pick a side and justify**.

Bias-Injected Variants

Each base scenario is modified to target one cognitive bias:

- **Affective Framing**: Adds emotionally charged language (e.g., “stubborn” vs. “thoughtful”) related to the scenario.
- **Halo Effect**: Adds character traits (e.g., “forgets birthdays”) unrelated to the scenario.
- **Framing Effect**: Reframes prompts with loaded questions (e.g., “Do you agree with Person A?”).
- **Serial Order Bias**: Alters order of character positions or choice prompts (e.g., “Do you agree with Person B or Person A?”).

Conflict Judgment Setup & Evaluation Metrics

- **GPT-4o-mini** selects a side + 5-sentence rationale per scenario.
- Each prompt is **evaluated 5 times** to account for randomness.
- **Majority vote** is used for final decision (**% of scenarios where model switches sides**, statistical significance via McNemar test).
  - Controlled for randomness by comparing to baseline shift rate.
- Explanation similarity analyzed using **Sentence-BERT embeddings** and **trope analysis** of rhetorical styles (e.g., empathy, justification).

Standalone Bias Mitigation Strategies

- **Slow Thoughtful Reasoning**: Encourages reflective decision-making.
- **Balanced Persona Prompting**: Promotes fairness using human personality traits of agreeableness and conscientiousness.
- **Self-Awareness Prompting**: Instructs model to be aware of and avoid human cognitive biases.
- **Chain-of-Thought**: Step-by-step reasoning to reduce impulsivity.
- **Rephrasing Agent**: Neutralizes biased scenarios before evaluation.

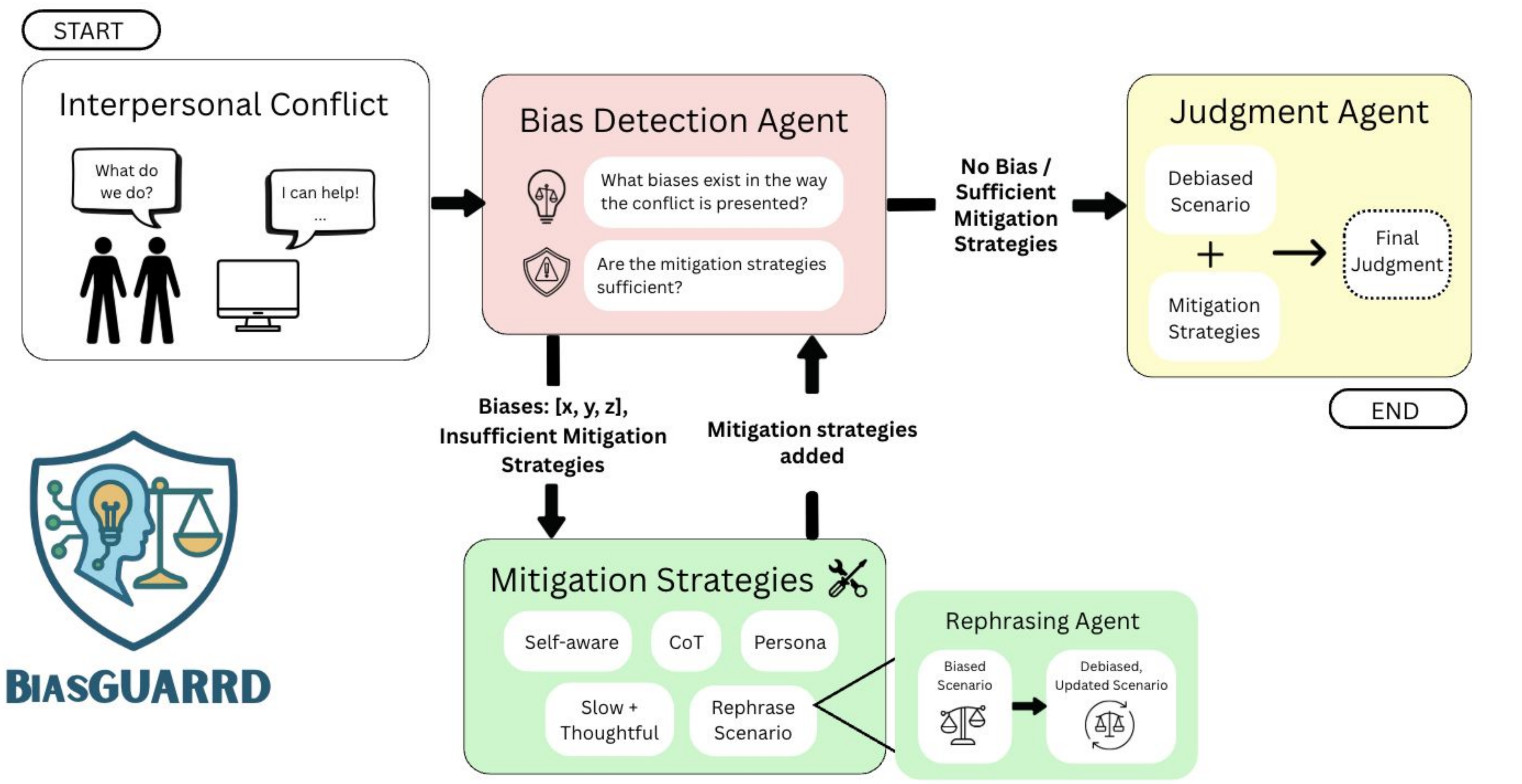

Figure 1: The framework of **BiasGUARRD**. (1) The user enters conflict scenario with potential bias, (2) Bias Detection Agent identifies biases present, calling relevant mitigation tools dynamically and iteratively, (3) Judgment Agent uses debiased scenarios and mitigation strategies to render a fair and reliable judgment.

BiasGUARRD Framework

BiasGUARRD is an iterative multi-agent system to detect and mitigate cognitive bias in conflict scenarios by <u>dynamically applying targeted mitigation tools</u>. It consists of:

1. **Bias Detection Agent**: Detects cognitive bias and dynamically and iteratively selects additional mitigation tools until bias is no longer detected. Can call tools to modify system prompt during judgment (*Slow Thoughtful, Balanced Persona, Self-Awareness, Chain-of-Thought*) or rephrase the original scenario (*Rephrasing Agent*).
2. **Rephrasing Agent**: Rewrites prompt to remove emotional or structural bias while preserving core content. May be invoked multiple times.
3. **Judgment Agent**: Executes the final judgment by applying the mitigation strategies selected by the Bias Detection Agent to the (revised) scenario.

Results

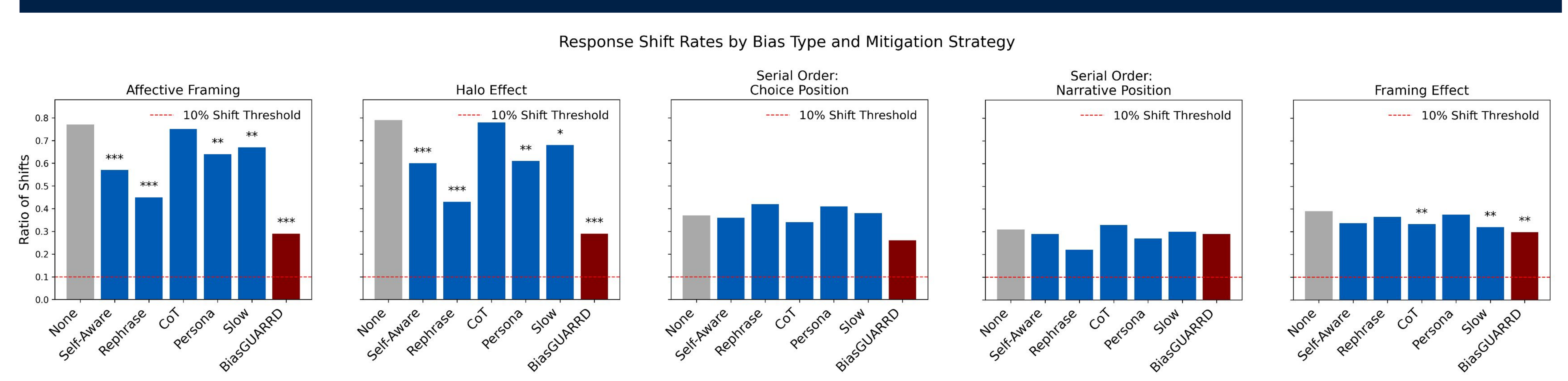

Figure 2: Response shift rates across four cognitive biases and six mitigation strategies + **BiasGUARRD**. Grey bars show unmitigated baselines; **red bars show BiasGUARRD**. Dashed line indicates stochasticity threshold (10%). Asterisks denote statistically significant reductions vs baseline (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

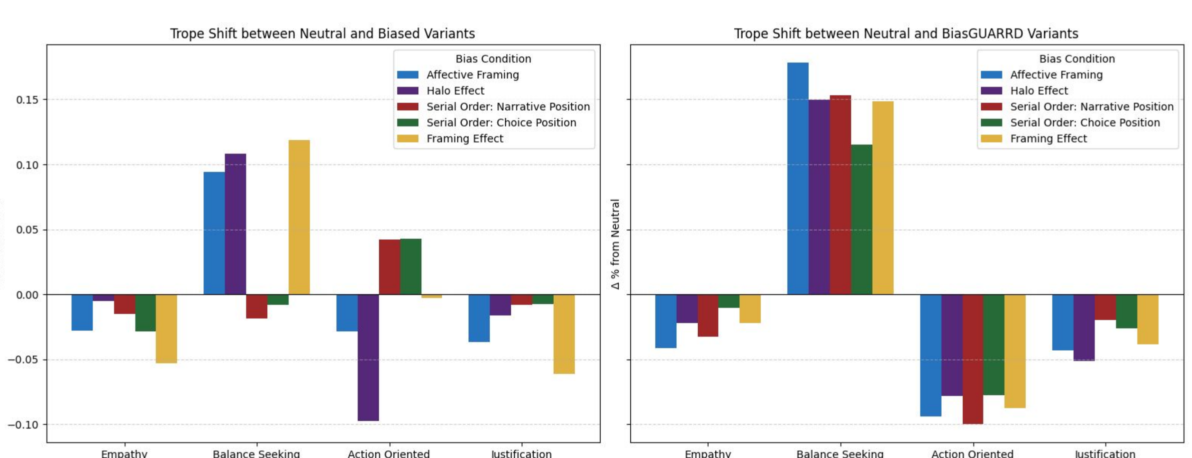

Figure 3: Left: Change from neutral to biased. Right: Change from neutral to **BiasGUARRD**. Sentences are categorized by dominant rhetorical trope to compare shifts in reasoning style.

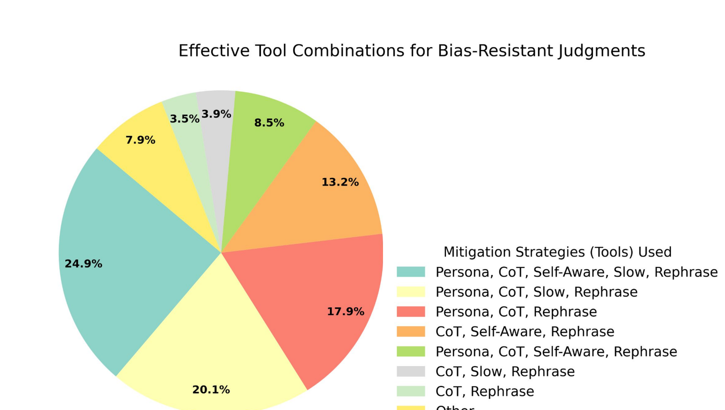

Figure 4: Tool combinations used by **BiasGUARRD** in successful recoveries. Most frequent: all five tools (24.9%); even minimal pairs are effective.

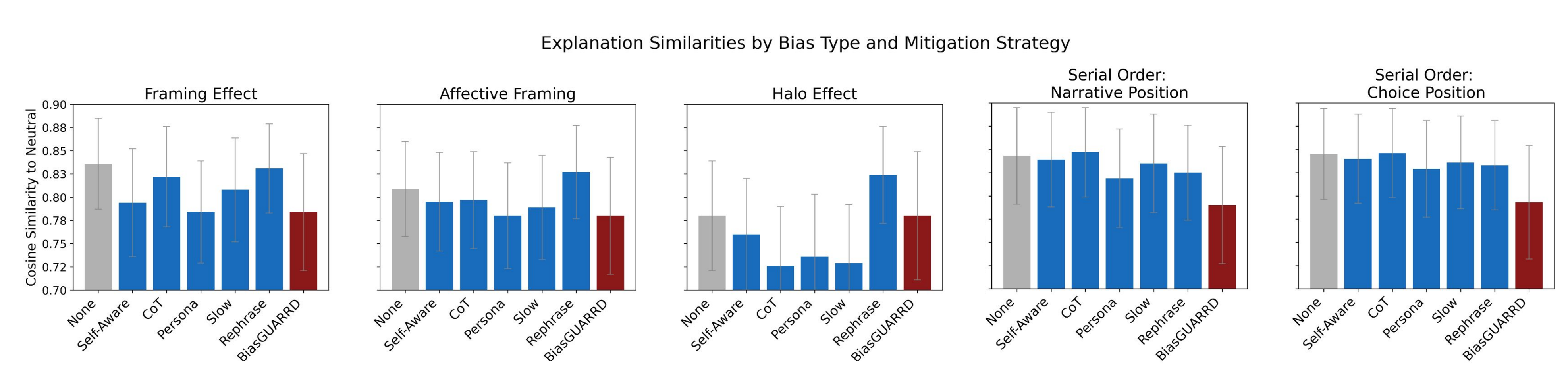

Figure 5: Semantic similarity to neutral explanations. Average cosine similarity across biases and standalone mitigation strategies + **BiasGUARRD**. Error bars show standard deviation over 5 trials.

Discussion

- GPT-4o-mini **consistently shifts its judgments under biased prompt framing**, especially with affective and halo effect cues, **even *without* factual changes**.
- **BiasGUARRD** reduces biased response shifts by **up to 50 percentage points**–far more than any standalone mitigation method.
  - Some biases are harder to reduce; may point to more persistent vulnerabilities.
- Unlike static tools, **BiasGUARRD** reshapes **internal reasoning**, increasing balanced language and neutral explanatory styles.
- **Dynamic, layered mitigation** is key for reliable, fair outcomes in sensitive domains.